

Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number : 3

Start date 18-September-25

Name: AKARAPU VASANTH

Internship Duration: 8 Weeks

1. Project Title

AI-Based Legal Reference and Case Retrieval System.

2. Project Objective

The primary goal of this internship project was to develop an AI-powered assistant capable of retrieving authoritative legal references and case law using natural language queries. The system aimed to simplify legal research for lawyers, paralegals, students, and the general public by providing concise, citation-backed answers and maintaining conversational context. This project aligns with the organization's mission to enhance legal accessibility and streamline research workflows through intelligent automation.

3. Project description in detail

The project involved building a **retrieval-augmented generation (RAG)** pipeline tailored to legal research using **Streamlit** as the user interface.

Key components included:

- **Document Ingestion:** Legal PDFs such as IPC, CrPC, and court judgments were extracted using **PyMuPDF**, cleaned, and segmented using a recursive character-based text splitter to maintain context.
- **Embedding & Indexing:** The text chunks were embedded using **Hugging Face's all-MiniLM-L6-v2** model and stored in a **Pinecone vector database** for fast and scalable semantic search.
- **Retrieval & Generation:** When a user submits a query, the system performs a similarity search in Pinecone to retrieve the most relevant legal sections. These are passed to **OpenAI's GPT model** with a legal reasoning prompt to generate concise, citation-backed responses.
- **UI & Interaction:** A **Streamlit-based conversational interface** allows users to ask questions, view references, manage chat history, and use guest login for quick access.
- **Memory & Feedback:** **Session state** preserves chat continuity, while user feedback helps evaluate and refine the accuracy of generated responses.

This system simplifies legal research by reducing manual search time, improving response accuracy, and making complex legal information more accessible through an intuitive AI-powered assistant.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Kickoff call, ChatGPT session, project setup, tech brush-up, share Milestone 1 tasks	Kickoff and setup calls completed; Milestone 1 tasks shared
Week 2	Begin Milestone 1: document collection, PDF loading, text normalization	Initial ingestion pipeline implemented; sample documents tested
Week 3	Milestone 1 guidance calls, continue preprocessing, validate chunking and metadata	Preprocessing refined; chunking and metadata scripts validated
Week 4	Complete Milestone 1, start Milestone 2: embedding and Pinecone setup	Milestone 1 completed; Pinecone index created and embedding pipeline started
Week 5	Milestone 2 guidance calls, test retrieval, refine index and chunk metadata	Retrieval tests run; index updated with metadata filters
Week 6	Complete Milestone 2, start Milestone 3: build RAG pipeline and Chainlit UI	Milestone 2 completed; RAG pipeline integrated with Chainlit UI
Week 7	Milestone 3 guidance calls, prompt tuning, provenance formatting, feedback integration	Prompt finalized; UI updated with provenance and feedback button
Week 8	Milestone 4: presentation prep, documentation, final review and closure call	Documentation completed; presentation prepared; project closure call held

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial orientation, ChatGPT session, project setup	18 September 2025
Prototype/First Draft	Chainlit UI with RAG pipeline and legal prompt integrated	15 October 2025
Mid-Term Review	Completed ingestion, indexing, and retrieval modules	3 November 2025
Final Submission	Documentation, presentation, and final system review	12 November 2025
Presentation	Final presentation and project closure call	13 November 2025

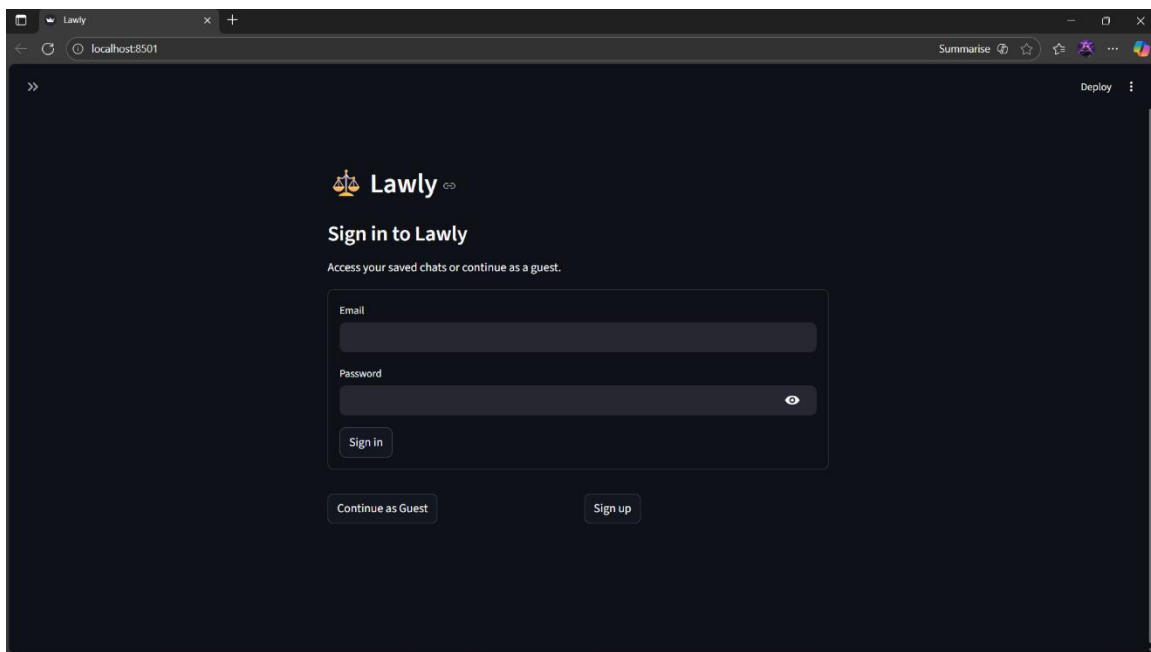
5b. Project execution details

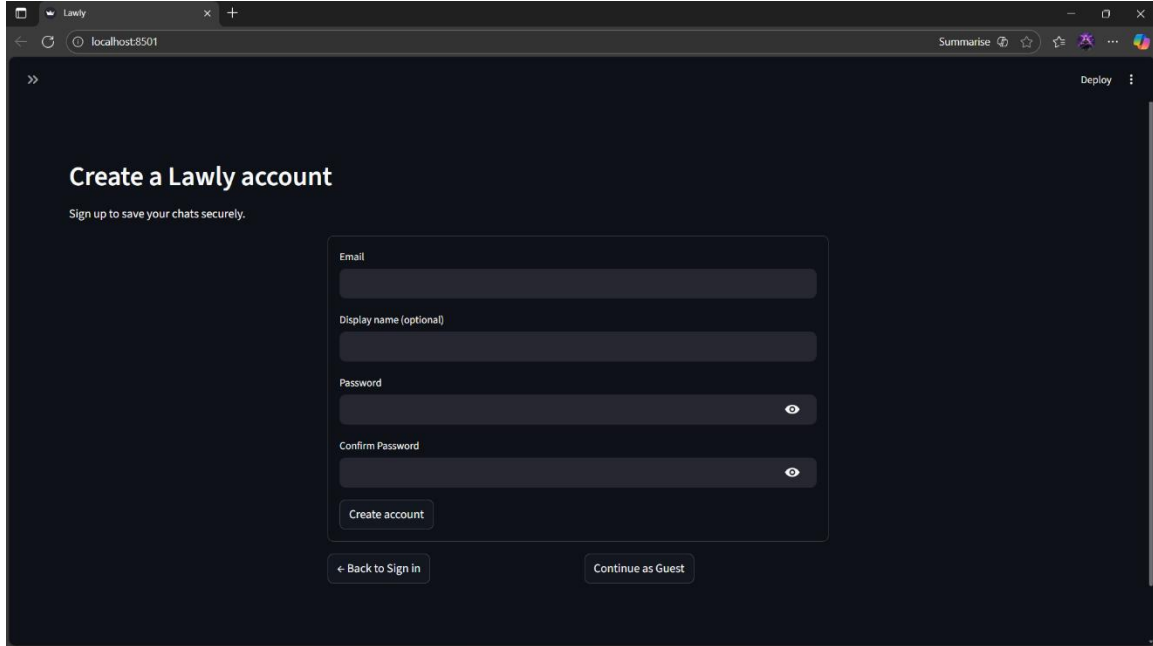
- **Weeks 1–2:** Collected legal documents such as IPC, CrPC, and judgments. Implemented **PDF extraction** using PyMuPDF and performed **text cleaning and normalization** for preprocessing.
- **Weeks 3–4:** Created text chunks and generated **semantic embeddings** using *Hugging Face's all-MiniLM-L6-v2* model. Populated a **Pinecone vector index** with metadata for efficient retrieval.
- **Weeks 5–6:** Developed the **RAG pipeline**, designed a domain-specific **legal reasoning prompt**, and integrated the system with a **Streamlit-based UI** for interactive query handling.
- **Weeks 7–8:** Conducted **testing, optimization, and bug fixes**, and prepared comprehensive **documentation** outlining the architecture, workflow, and evaluation methods.

Each phase incorporated **validation scripts, modular utilities, and reusable components**, ensuring reproducibility, scalability, and maintainability of the entire system.

6. Snapshots / Screenshots

Include





A screenshot of a web browser showing the 'Create a Lawly account' page. The browser's address bar shows 'localhost:8501'. The page has a dark theme. At the top, there's a 'Deploy' button. The main heading is 'Create a Lawly account' with a subtext 'Sign up to save your chats securely.' Below this is a form with four input fields: 'Email', 'Display name (optional)', 'Password', and 'Confirm Password'. Each field has a corresponding eye icon for toggling visibility. At the bottom of the form is a 'Create account' button. Below the form are two buttons: '← Back to Sign in' and 'Continue as Guest'.

Create a Lawly account

Sign up to save your chats securely.

Email

Display name (optional)

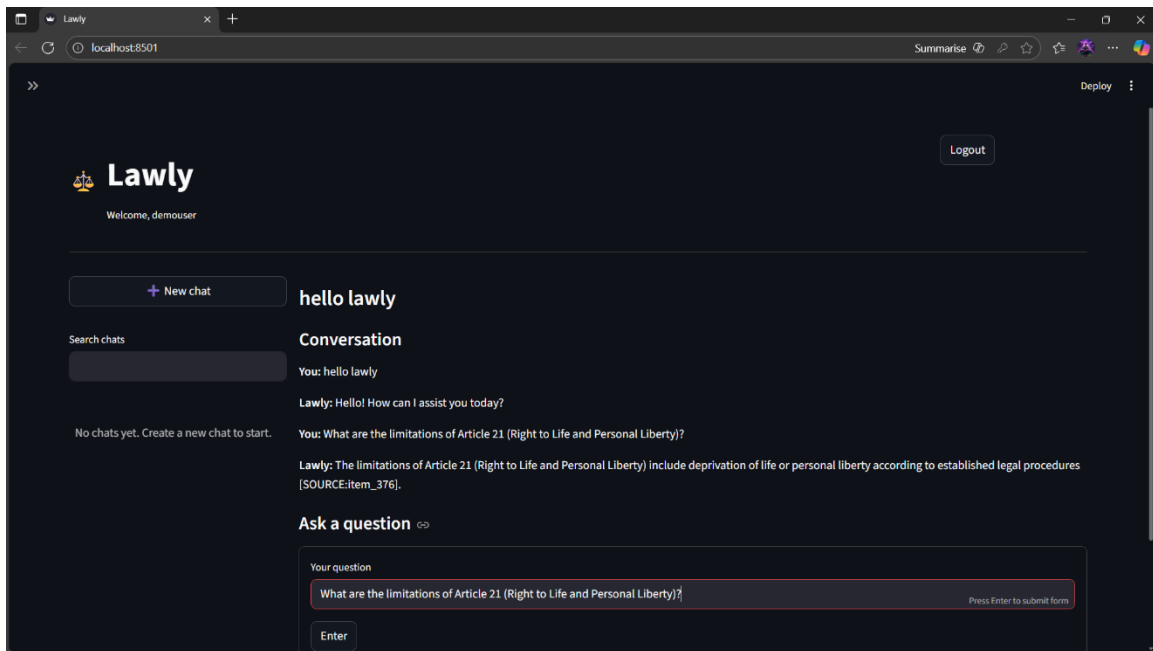
Password

Confirm Password

Create account

← Back to Sign in

Continue as Guest



A screenshot of the Lawly chat interface. The browser's address bar shows 'localhost:8501'. The page has a dark theme. At the top, there's a 'Logout' button. The main heading is 'Lawly' with a subtitle 'Welcome, demouser'. Below this is a 'New chat' button. To the left of the chat area is a 'Search chats' input field. The chat area shows a conversation with 'hello lawly'. The conversation starts with 'You: hello lawly', followed by 'Lawly: Hello! How can I assist you today?', and then 'You: What are the limitations of Article 21 (Right to Life and Personal Liberty)?'. The response from Lawly is 'The limitations of Article 21 (Right to Life and Personal Liberty) include deprivation of life or personal liberty according to established legal procedures [SOURCE:item_376]'. At the bottom, there's an 'Ask a question' section with a text input field containing 'What are the limitations of Article 21 (Right to Life and Personal Liberty)?' and a 'Press Enter to submit form' button. Below the input field is an 'Enter' button.

Logout

Lawly

Welcome, demouser

New chat

Search chats

hello lawly

Conversation

You: hello lawly

Lawly: Hello! How can I assist you today?

You: What are the limitations of Article 21 (Right to Life and Personal Liberty)?

Lawly: The limitations of Article 21 (Right to Life and Personal Liberty) include deprivation of life or personal liberty according to established legal procedures [SOURCE:item_376].

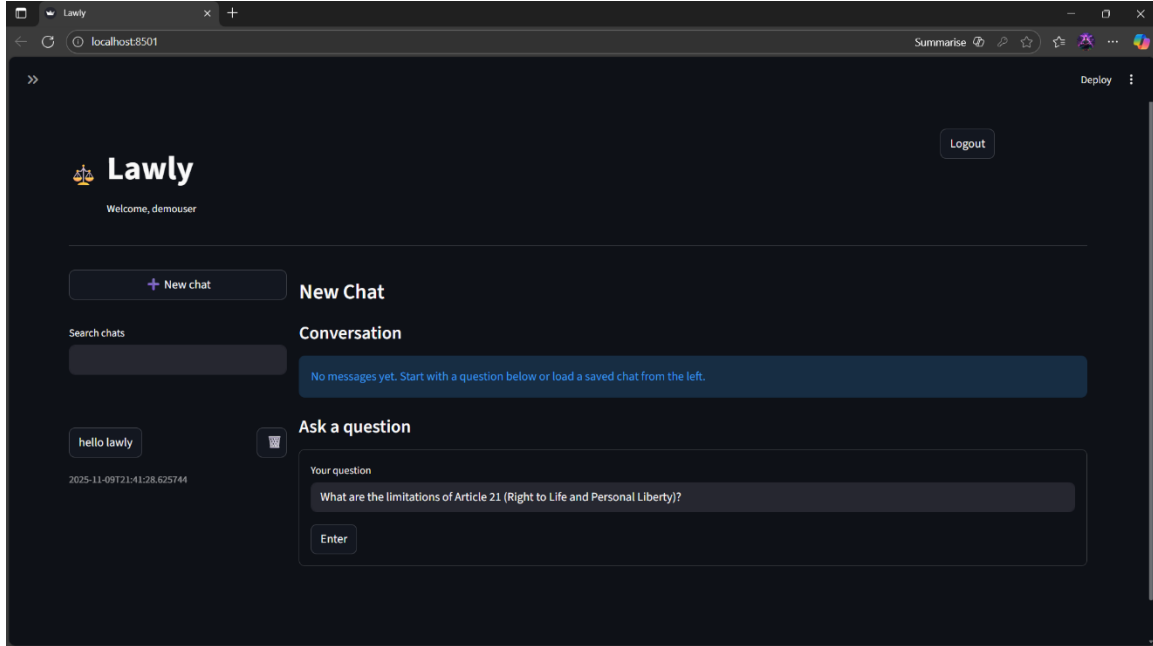
Ask a question

Your question

What are the limitations of Article 21 (Right to Life and Personal Liberty)?

Press Enter to submit form

Enter



7. Challenges Faced

Embedding Dimension Mismatch: Initial errors due to incompatible embedding model and Pinecone index dimensions were resolved by aligning model output with index configuration.

- **Region Compatibility:** Pinecone region setup required troubleshooting due to API key and environment conflicts.
- **Metadata Extraction:** Parsing court metadata from judgments was inconsistent; resolved by refining regex patterns and fallback logic.
- **UI Provenance Display:** Ensuring chunk-level citation visibility in Chainlit required custom formatting and prompt tuning.

These challenges were mitigated through persistent debugging, documentation reviews, and iterative testing.

8. Learnings & Skills Acquired

- **Technical Skills:**
 - Semantic search pipeline design (chunking, embedding, indexing)
 - Pinecone integration and metadata filtering
 - RAG architecture and LLM prompting
 - Chainlit UI prototyping and conversational memory
- **Soft Skills:**

- Problem-solving under ambiguity
- Clear documentation and reproducible workflows
- Communication of technical decisions and trade-offs
- **Domain Knowledge:**
- Legal document structure and terminology
- Statutory interpretation and judgment referencing

9. Testimonials from team

This project was a rewarding experience. I successfully built a functional legal assistant prototype that impressed mentors with its clarity, modularity, and real-world relevance. The feedback highlighted my persistence in debugging, my attention to provenance and auditability, and my ability to translate legal requirements into actionable technical solutions.

10. Conclusion

This internship provided a deep dive into building scalable NLP systems for legal tech. It strengthened my backend engineering skills and broadened my understanding of retrieval systems and conversational AI. The project aligns perfectly with my academic goals and career aspirations in legal AI and semantic search, and it serves as a strong portfolio piece for future opportunities.

11. Acknowledgements

I sincerely thank my mentor and the organization for their guidance and support throughout this internship. Special thanks to the engineering team for their feedback and collaboration, and to the legal advisors who helped validate the system's relevance and accuracy.